

$H = \$1$ million and she also has financial wealth of $W = \$1$ million. Her total wealth is thus $H+W = \$2$ million. Like our first investor, we'll assume her risk aversion is $\gamma = 5$.

The optimal holding of risky equities with risk-free labor income is:

$$w^* = \frac{1}{\gamma} \frac{\mu - r_f}{\sigma^2} \left(1 + \frac{H}{W} \right). \quad (5.2)$$

Note that when human capital is zero, $H = 0$, equation (5.2) is exactly the same as equation (5.1). Our investor with risk-free human capital tilts her portfolio toward stocks relative to an investor with only financial wealth. The proportion of equities held by the investor in this example is:

$$w^* = \frac{1}{\gamma} \frac{\mu - r_f}{\sigma^2} \left(1 + \frac{H}{W} \right) = \frac{1}{5} \frac{0.10 - 0.02}{(0.15)^2} \left(1 + \frac{1}{1} \right) = 125\% \quad (5.3)$$

The optimal holdings of our investor with labor income are:

$$\begin{aligned} \text{Hold } 1.25 \times \$1 \text{ million} &= \$1,250,000 \text{ in equities} \\ &\quad - \underline{\$250,000 \text{ in T-bills}} \\ &\quad \$1 \text{ million} \end{aligned}$$

This is a much larger weight in equities than our first investor without labor income; the investor with risk-free labor income shorts the risk-free asset and holds a levered portfolio of 125% in equities. But it turns out that the combined portfolio has exactly the same 62.5% holdings in risky assets.

If we view the investor's *total wealth* portfolio, we have:

$$\begin{aligned} \text{Endowed with} &\quad \$1,000,000 \text{ in human capital} \\ \text{Hold } 1.25 \times \$1 \text{ million} &= \$1,250,000 \text{ in equities} \\ &\quad - \underline{\$250,000 \text{ in T-bills}} \\ &\quad \$2 \text{ million} \end{aligned}$$

As a fraction of total wealth, her equity allocation is $1,250,000/2 \text{ million} = 62.5\%$, which is the same as our first investor without labor income.

The presence of risk-free labor income allows the investor to take on more risk in her financial portfolio. In fact, equation (5.2) levers up the standard equity position (from equation (5.1)), and the leverage ratio is the ratio of human capital to financial wealth, H/W .

Asset allocation must be viewed in the broader context of total wealth, which sums both financial and human capital wealth. Risk-free labor income is equivalent to being endowed with risk-free bonds. Your financial portfolio has to hold more equities to balance out your risk-free human capital. In economists' jargon, the risk exposure in human capital wealth *crowds out* the risk-taking ability of our financial wealth.